

Optimizing VQE Time: Impact of Hamiltonian Truncation

Motivation & Approach

- Quantum simulations suffer from exponential computational scaling.
- Evaluating the Hamiltonian expectation value ($\langle \psi | H | \psi \rangle$) dominates the total simulation runtime.
- Systematically pruning low-coefficient Pauli terms (below threshold Th_1) streamlines the Hamiltonian and accelerates evaluation times.

Hamiltonian Chopping

Qiskit's mapping pipeline is intercepted to discard insignificant terms prior to VQE evaluation.

```
def truncated_map(second_q_op):
    qubit_op = original_map(second_q_op)
    # Recursively prune mapped Pauli operators below the static threshold
    if isinstance(qubit_op, dict):
        return {k: SparsePauliOp(v).chop(self.ham_threshold) if v is not None else None for k, v in
qubit_op.items()}
    elif isinstance(qubit_op, list):
        return [SparsePauliOp(v).chop(self.ham_threshold) if v is not None else None for v in
qubit_op]
    elif qubit_op is not None:
        return SparsePauliOp(qubit_op).chop(self.ham_threshold)
```

Experimental Configuration

- The target system is Hydrogen (H_2) at a 0.74 Å bond length.
- Classical preprocessing utilizes PySCF with a 6-31G* basis set and a 2-electron, 2-orbital (2e, 2o) active space.
- The quantum circuit applies a Jordan-Wigner mapping and a UCCSD Ansatz.

Unchopped Hamiltonian Landscape

Pauli String	Coefficient	Pauli String	Coefficient
IIII	-0.99608525	IIZZ	0.08956863
IZIZ	0.16015921	ZIII	0.04395566
IZII	0.12314796	IIZI	0.04395566
IIIZ	0.12314796	XXXX	0.02205342
IZZI	0.11162205	YYXX	0.02205342
ZIIZ	0.11162205	XXYY	0.02205342
ZIZI	0.09933742	YYYY	0.02205342
ZZII	0.08956863		

The baseline Hamiltonian consists of 15 total terms, with a maximum absolute coefficient of 0.996 and a minimum of 0.022.

Truncation Performance Metrics

Threshold (Th1)	Remaining Terms	Term Reduction	Execution Time (s)	Abs. Delta E (Ha)
0.000 (Baseline)	15	0.00%	0.5748	0.00000000
0.110	6	60.00%	0.4807	0.01479744
0.120	4	73.33%	0.3743	0.23804151
0.140	2	86.67%	0.9883	0.16401904
0.160	2	86.67%	1.0952	0.16401904
0.175	1	93.33%	0.3097	0.32417824

Performance Analysis & Limitations

- Reducing the term count from 15 to 4 yields a clear reduction in execution time, validating the slimming strategy.
- Extreme truncation destroys the physical model, which heavily distorts the energy landscape, confuses the VQE, and causes severe execution time spikes.
- Hamiltonian truncation is highly effective, but extreme pruning on small toy models introduces mathematical instability.

